

NAGALAND LITERACY AND NUMERACY FEST 2026

Grade 8 · Micro-Improvement Project (Numeracy Teacher) Grade 8

Chapter 8: We Distribute, Yet Things Multiply

Why Micro-Improvement Project (MIP)?

In Nagaland, distribution and pattern are part of daily life long before they appear in a textbook. A weaver preparing a Tsungkoteptsu or a Lohe shawl repeats the same coloured strip across every panel; a mother dividing rice or king chilli equally among family members is distributing a quantity across parts; a farmer laying out a paddy terrace is combining two boundaries into one field. In Grade 6 (Chapter 6: Perimeter and Area), students learned to deduce formulas for boundaries and surfaces through pattern and generalisation. In Grade 8 (Chapter 8: We Distribute, Yet Things Multiply), students extend that same instinct noticing a pattern and generalising a rule from shapes to algebraic expressions: distributing multiplication over addition and subtraction, expanding brackets, multiplying binomials, and discovering identities that make mental mathematics faster and more confident.

Skill in Focus	Distributive Property · Multiplication of Algebraic Expressions · Binomial Multiplication · Algebraic Identities · Pattern Recognition · Multiple Solution Strategies
Duration	1 Week (35 minutes)

Learning Outcomes

Theme	Key Concepts
Distributive Property	Understanding multiplication over addition and subtraction: $a(b+c) = ab+ac$ and $a(b-c) = ab-ac$.
Multiplication of Algebraic Expressions	Expanding expressions involving variables using the distributive law.

Learning Objective: Grade 8 Learning Objectives

1. Explain the distributive property of multiplication over addition and subtraction.
2. Expand algebraic expressions using the distributive law accurately.
3. Multiply two algebraic expressions involving variables.
4. Solve real-life and mathematical problems using the distributive property and algebraic identities.

Teachers may adapt activities to their own school context. These are only suggestive.

Detailed Tasks

Task-1

Observe how students perceive distribution and multiplication in daily classroom routines.

Before teaching, find out what each student already knows about sharing, distributing, and combining quantities. You do not need any extra time or materials to watch and listen during normal classroom and school activities.

1. While distributing notebooks or chalk, ask: 'If I give 3 pieces of chalk to each of 5 rows, how many pieces do I need in total — and is there more than one way to count it?'
2. At break, ask: 'If we have 8 oranges and 4 king chillies to share equally between 2 friends, how would you work it out quickly?'
3. During the games period, ask: 'If each of the 6 teams needs 2 long ropes and 1 short rope, how many ropes altogether?'
4. Note who breaks the problem into parts before combining, who guesses, and who needs the question repeated.

Student Name	Breaks quantity into parts	Combines parts correctly	Uses multiplication language	Needs support

Scale: ☐ Not Yet ☐ Sometimes ☐ Consistently

Task-2

Plan classroom and outdoor activity routine using familiar Nagaland contexts.

A predictable routine across the three weeks helps students connect distribution, expansion, and identities to everyday experience.

1. Fix six 40-minute periods across three weeks in the timetable, one per activity below, plus daily 10–15 minute homework.
2. Use familiar contexts such as bamboo basket-weaving, kitchen garden plots, paddy field terraces, Naga shawl looms, and the local market.
3. Include mathematical talk by using words like distribute, expand, term, product, factor, and identity in every session.
4. Balance indoor algebra work with outdoor measurement-based activities wherever the school compound allows it (see indoor/outdoor tags in Task-4).
5. Ensure every student gets a hands-on turn — with materials, on the board, or in group presentation.

Task-3

Arrange materials and classroom setup to support algebra visualisation.

Hands-on materials help students see distribution, expansion, and identities physically before working with symbols alone. Ask students to bring a few bamboo strips, some string, and old newspaper, and keep them in a corner of the classroom; this becomes the class's 'algebra kit' for the three weeks.

1. Use simple, locally available materials only, nothing needs to be purchased specially.
2. Focus on students arranging, measuring, colouring, and building rather than only reading formulas from the board.
3. Keep the kit in a common place so all students can access it easily during group work.
4. Rotate who collects and returns materials each session so responsibility is shared.

Bamboo Strips / Sticks	String / Rope	Grid or Plain Paper	Colour Pens / Crayons
Chalk	Old Newspaper (for cutting)	Scale / Ruler	Pebbles or Seeds (for counting)

Task-4

Conduct classroom (and select outdoor) activities that build distributive reasoning, expansion, and identities.

Every student must participate in every session. Keep activities short, hands-on, and connected to something they already know from Nagaland life. The goal is understanding students should be able to explain what they distributed, expanded, or matched, and why, not just state an answer.

Suggested Activities (teachers can modify or add more as per their context)

Activity 1: Sharing the Bamboo Bundle

Concept built: Distributive property: $a(b+c) = ab+ac$ and $a(b-c) = ab-ac$

Setting: Indoor (Classroom)

Materials Required (low-cost, locally available): bamboo strips or sticks (or pencils as a substitute), chalk, notebook

Objective of the Task: Students understand the distributive property by physically distributing bundled bamboo strips into equal groups before multiplying, then comparing the total with multiplying the whole bundle at once.

Teacher's Note: Open with a weaver preparing baskets: 5 baskets, each needing 3 long strips and 2 short strips. Ask students to find the total two ways basket by basket, or all long strips then all short strips and show both give the same answer: $5(3+2) = 5 \times 3 + 5 \times 2$. Let students physically arrange strips into groups rather than only calculating. Watch for the common error of writing $a(b+c)$ as $ab+c$, and correct it by pointing back to the physical groups. Introduce the subtraction case using a basket with a few strips removed (e.g., damaged or reused elsewhere).

Time & Classroom Activity

Time	Activity
5 min	Hook: the story of a weaver preparing bamboo strips for several baskets.
10 min	Teacher models on the board: 5 groups of (3 long + 2 short) strips, computed two ways, written as $5(3+2) = 5 \times 3 + 5 \times 2$.
15 min	Group activity: each group gets a different scenario (e.g., 4 kitchen gardens each needing 6 bamboo pegs and 3 stones); they physically arrange strips/pebbles and write both the expanded and factored form.
7 min	The teacher introduces the subtraction case $a(b-c)$, e.g., 'removing spoilt oranges from baskets', and works one example with the class.
3 min	Share out: two groups present their scenario and expression to the class.

Homework (Segregated from Classroom Activity): Students find one real distribution example at home (e.g., a parent dividing rice, vegetables, or firewood among family members), write it as $a(b+c)$ or $a(b-c)$, and solve it numerically in their notebook (10–15 minutes).

Activity 2: Expanding the Garden Plot

Concept built: Multiplication of algebraic expressions

Setting: Outdoor (school garden or field) — with an indoor grid-paper alternative for rainy days

Materials Required (low-cost, locally available): string/rope, wooden pegs or sticks, chalk, measuring tape or hand-spans, notebook (indoor alternative: grid paper, scale, colour pens)

Objective of the Task: Students expand algebraic expressions with variables (a monomial multiplied by a binomial or longer expression) using the distributive law, connecting the expression to the area of a real or sketched garden plot.

Teacher's Note: Take the class to a marked garden plot outdoors. Represent an unknown length using a string of length ' x hand-spans', and tie on an extra string of length ' 3 hand-spans' to show a total width of $(x+3)$. Multiply this by a depth of ' $2x$ ' to build the area expression $2x(x+3) = 2x^2 + 6x$, verifying by measuring physically. If the weather or space does not allow going outdoors, do the identical activity indoors by sketching plots on grid paper and mention both options to students so the mathematics is not lost to weather.

Time & Classroom Activity

Time	Activity
5 min	Recap the distributive property from Activity 1.
10 min	The teacher demonstrates on the outdoor plot (or grid sketch): building the expression $2x(x+3)$ and expanding it.
15 min	Group activity: each group gets a different plot expression (e.g., $3x(2x+5)$, $4y(y-3)$), marks it out with string/pegs or sketches it on grid paper, expands it, and verifies by substituting a chosen number for the variable.
7 min	Whole-class check: compare two groups' work on the board and discuss any common errors.
3 min	Wrap-up: connect the activity to planning a real kitchen garden at home.

Homework (Segregated from Classroom Activity): Expand four algebraic expressions set by the teacher in the notebook, and sketch one of them as a small plot diagram (15 minutes).

Activity 3: Two Paddy Fields, One Product

Concept built: Binomial multiplication: $(a+b)(c+d)$

Setting: Indoor (Classroom)

Materials Required (low-cost, locally available): grid or plain paper, colour pens, scale, scissors (optional)

Objective of the Task: Students multiply two binomials using an area model, representing two adjoining paddy terraces with different lengths and breadths that are combined into one field.

Teacher's Note: Paddy fields in Nagaland are often terraced, with adjoining plots belonging to different families a natural real-life picture for combining two binomials. Draw a rectangle divided into four smaller regions representing the terms ac , ad , bc , and bd , and walk through one worked example, e.g., $(x+2)(x+5)$, on the board before groups attempt their own. Encourage students to label every region clearly and to check their expanded answer by substituting numbers for the variables.

Time & Classroom Activity

Time	Activity
5 min	Recap expanding a single term times a binomial (Activity 2).
10 min	Teacher models $(a+b)(c+d)$ as the area of a combined rectangle, drawing and labelling all four regions.
15 min	Group activity: each group gets its own binomial pair (e.g., $(x+2)(x+5)$, $(2a+3)(a+4)$), draws the area model, expands it, and verifies numerically.
7 min	Share out: two groups present their diagrams and expanded expressions.
3 min	Wrap-up: connect back to the real terrace-field picture and preview identities coming next.

Homework (Segregated from Classroom Activity): Multiply three binomial pairs set by the teacher using a small area-model sketch in the notebook (15 minutes).

Activity 4: Shawl Block Identities

Concept built: Algebraic identities: $(a+b)^2$, $(a-b)^2$, $(a+b)(a-b) = a^2-b^2$

Setting: Indoor (Classroom)

Materials Required (low-cost, locally available): plain or coloured paper, pencil, scale, colour pens or crayons

Objective of the Task: Students discover and apply the three standard identities by designing coloured Naga shawl block patterns, where each coloured section represents one algebraic term.

Teacher's Note: Begin with a short discussion of Naga shawl patterns. Students already know the Tsungkotepsu of the Ao, the Lohe of the Sumi, the Supong and point out that repeating geometric blocks already 'hide' algebra inside them. On the board, draw a square of side $(a+b)$, divide it into four sections (a^2 , ab , ab , b^2), and write $(a+b)^2 = a^2+2ab+b^2$. For $(a-b)^2$, show the same shape with the $2ab$ section shaded differently to show it is subtracted. For a^2-b^2 , draw a large square with a smaller square removed from one corner. Encourage students to name their own design after their village or tribe's shawl motif.

Time & Classroom Activity

Time	Activity
5 min	Hook: brief discussion of Naga shawl patterns students already recognise.
10 min	Teacher models $(a+b)^2$ as a shawl block square divided into four labelled, coloured sections.
15 min	Group activity: each group is assigned one identity — $(a+b)^2$, $(a-b)^2$, or a^2-b^2 — and designs a coloured shawl block for it, labelling every section and writing the identity underneath.
7 min	Mini gallery share: two groups display their block and explain the connection to the algebra.
3 min	Wrap-up: bring all three identities together on the board and preview their use for quick mental mathematics.

Homework (Segregated from Classroom Activity): Design the shawl block for one identity the student's group did not cover in class, label all sections, and write the identity below the design (15–20 minutes).

Activity 5: Market Shortcut Mathematics

Concept built: Mental mathematics using identities

Setting: Indoor (Classroom) — role-played as a local market stall

Materials Required (low-cost, locally available): no special materials needed; optional price cards or paper slips with local item names (oranges, king chilli, bamboo shoots)

Objective of the Task: Students use algebraic identities to perform quick mental calculations, such as finding 102×98 using $(100+2)(100-2) = 100^2 - 2^2$, instead of long multiplication.

Teacher's Note: Frame the challenge as a shopkeeper at a local market who needs to calculate quantities and prices quickly for example, '98 baskets of king chilli at ₹102 each.' Demonstrate 3–4 identity-based shortcuts on the board before students create their own. Encourage students to first estimate mentally, then check with the standard method, so they see the time saved and gain confidence in the shortcut.

Time & Classroom Activity

Time	Activity
5 min	Quick challenge: who can solve the 102×98 fastest standard method vs identity shortcut?
10 min	The teacher demonstrates three to four identity-based shortcuts using local market-style numbers.
15 min	Pair activity: students create their own market-price shortcut problems using round numbers, swap with a partner, solve using an identity, and verify with the standard method.
5 min	Whole-class share: the fastest or most creative shortcut found by any pair.

Homework (Segregated from Classroom Activity): Solve five shortcut problems set by the teacher using identities, showing which identity was used for each (10 minutes).

Activity 6: Weaving the Pattern Rule

Concept built: Pattern recognition and multiple solution strategies (chapter synthesis)

Setting: Indoor (Classroom)

Materials Required (low-cost, locally available): grid paper, colour pens, ruler; local woven-pattern samples or photographs if available

Objective of the Task: Students identify a numerical or algebraic pattern inspired by a repeating woven design, then solve one combined problem using more than one valid strategy and justify why all methods agree.

Teacher's Note: Show or describe a repeating woven pattern from a local textile (stripes, diamonds, zig-zags) and ask students to describe the 'rule' behind the repetition, linking it to the pattern-building done all through the chapter. Then pose one synthesis problem that combines distribution, expansion, and an identity, and assign different groups a different valid strategy direct expansion, identity shortcut, or area model so the class can compare methods and discuss why they all give the same answer. This activity works well as a review near the end of the chapter.

Time & Classroom Activity

Time	Activity
5 min	Warm-up: spotting the rule in a local woven pattern (stripes, diamonds, or similar).
10 min	The teacher poses one synthesis problem and briefly reviews three possible solution strategies.
15 min	Group activity: each group solves the same problem using its assigned strategy and records every step.
7 min	Cross-group comparison: groups present their method; the class discusses why all three agree.
3 min	Wrap-up: connect pattern-thinking in weaving to pattern-thinking in algebra.

Homework (Segregated from Classroom Activity): Solve one problem set by the teacher using two different methods, and write one sentence explaining why both give the same answer (15 minutes).

Task-5

Reflect on students' understanding and application of distribution, expansion, and identities.

Observe 5–6 students each week during activities across the three weeks. No written test is required, though a short optional quiz may be used if helpful. Watch what students do with materials and listen to how they explain their thinking. Encourage progress, not perfection.

Student Name	Applies distributive law correctly	Expands expressions accurately	Recognises/applies identities	Explains reasoning clearly

Student Name	Applies distributive law correctly	Expands expressions accurately	Recognises/applies identities	Explains reasoning clearly

Scale: ☐ Not Yet ☐ Sometimes ☐ Consistently

Chapter Topic Coverage

Chapter Topic	Covered in Project?	Activity
Distributive property	Yes	Activity 1 — Sharing the Bamboo Bundle
Multiplication of algebraic expressions	Yes	Activity 2 — Expanding the Garden Plot
Binomial multiplication	Yes	Activity 3 — Two Paddy Fields, One Product
Algebraic identities: $(a+b)^2$, $(a-b)^2$, a^2-b^2	Yes	Activity 4 — Shawl Block Identities
Mental mathematics using identities	Yes	Activity 5 — Market Shortcut Mathematics
Pattern recognition & multiple solution strategies	Yes	Activity 6 — Weaving the Pattern Rule

Classroom Project Cycle

In the previous cycle (Grade 6, Chapter 6: Perimeter and Area), you focused on:

1. Helping students understand perimeter as the distance around a shape, and area as the surface it covers.
2. Guiding students to deduce formulas through pattern and generalisation rather than by memorising a rule.
3. Using familiar Nagaland contexts paddy terraces, bamboo fences, courtyards, and football grounds to connect formulas to daily life.

In this cycle (Grade 8, Chapter 6: We Distribute, Yet Things Multiply), you will:

1. Help students see distribution, expansion, and multiplication of algebraic expressions as extensions of everyday sharing and combining.

2. Guide students to discover algebraic identities through visual, hands-on shawl-block and area-model designs rather than by memorising formulas.
3. Build mental mathematics confidence by connecting identities to quick, real-life shortcuts.
4. Use familiar Nagaland contexts bamboo weaving, kitchen gardens, paddy terraces, Naga shawl motifs, and the local market to root algebra in daily life.

BY THE END OF THIS PROJECT, TEACHERS ALONG WITH STUDENTS ARE REQUIRED TO COME UP WITH INNOVATIVE PROJECTS WHICH CAN BE EXHIBITED WITHIN THE CLASSROOM.

For example, groups may combine their shawl-block identity designs, garden-plot expansions, and market shortcut challenge cards into a small classroom exhibition 'Algebra in Nagaland Life' modelled on the gallery-walk style used in earlier cycles, where every visitor can see the design and understand the mathematics behind it.

Picture Guide



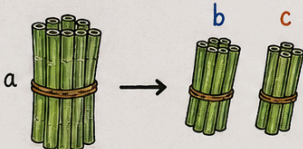
WE DISTRIBUTE, YET THINGS MULTIPLY



Grade 8 - Chapter 8

THEME: Distribution, Patterns and Algebra in our daily life

1. DISTRIBUTIVE PROPERTY

$$a(b+c) = ab + ac$$


$$3(4+2) = 3 \times 4 + 3 \times 2 = 12 + 6 = 18$$

Sharing bamboo bundles

2. EXPANDING EXPRESSIONS

$$3(x+2) = 3x + 6$$


Expand using distributive property

3. MULTIPLY BINOMIALS

$$(x+a)(x+b)$$

	x	b
x	x ²	bx
a	ax	ab

$$(x+a)(x+b) = x^2 + bx + ax + ab$$

Multiply two binomials

4. ALGEBRAIC IDENTITIES

	x	2	
x	x ²	2x	$(x+2)^2 = x^2 + 4x + 4$
2	2x	4	

Shortcut to solve fast

USEFUL IDENTITIES

$(a+b)^2 = a^2 + 2ab + b^2$

a ²	ab
ab	b ²

$(a-b)^2 = a^2 - 2ab + b^2$

a ²	-ab
-ab	b ²

IN OUR DAILY LIFE






Sharing equally

Repeating pattern

Field layout

Market distribution

OUR TAKEAWAY

- ✓ Distribute to simplify.
- ✓ Expand to understand.
- ✓ Patterns help us generalise.
- ✓ Identities make maths easy.
- ✓ Maths is in our daily life!



THINK PATTERN • USE DISTRIBUTION • SOLVE SMART